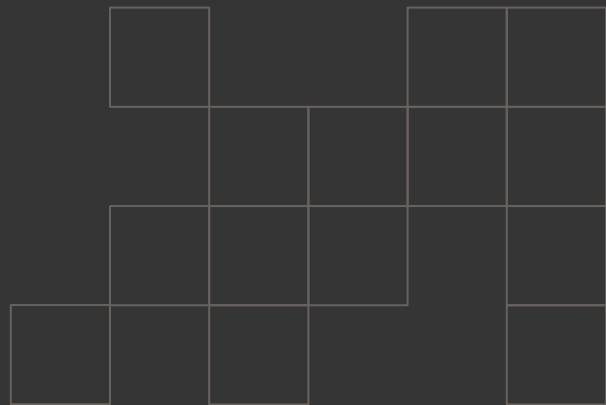


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03/25/202

Temperature Forecasting with Nixtla



Contents

1.

Overview

2.

Data Prep

3.

Model Selection & Training

4.

Feature Engineering

5 & 6.

Model Evaluation and Validation

7.

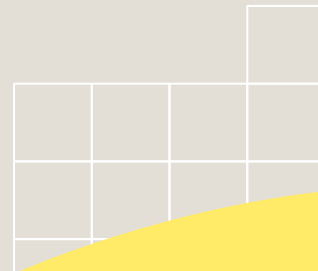
Generative Modeling

8.

Impact of Generative Models

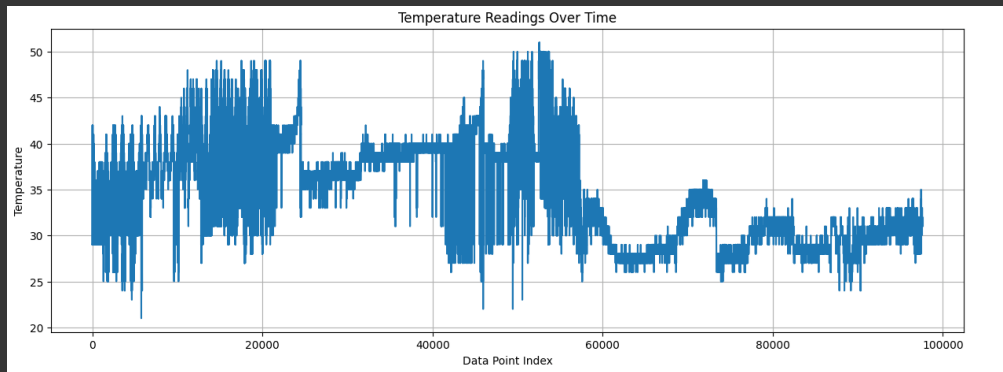
9.

Challenges, Insights, Takeaways



Overview

- Temperature forecasting using Nixtla TimeGPT
- Goal: Predict IoT sensor temperatures
- Dataset: Kaggle IoT temperature readings



Data Prep

- Loaded & explored dataset
- Converted datetime, indexed, sorted
- Resampled to hourly frequency
- No missing values

	id	room_id/id	noted_date	temp	out/in
0	__export__.temp_log_196134_bd201015	Room Admin	08-12-2018 09:30	29	In
1	__export__.temp_log_196131_7bca51bc	Room Admin	08-12-2018 09:30	29	In
2	__export__.temp_log_196127_522915e3	Room Admin	08-12-2018 09:29	41	Out
3	__export__.temp_log_196128_be0919cf	Room Admin	08-12-2018 09:29	41	Out
4	__export__.temp_log_196126_d30b72fb	Room Admin	08-12-2018 09:29	31	In

	temp
count	97606.000000
mean	35.053931
std	5.699825
min	21.000000
25%	30.000000
50%	35.000000
75%	40.000000
max	51.000000

```
df.isnull().sum()
```

	0
id	0
room_id/id	0
noted_date	0
temp	0
out/in	0

dtype: int64

Model Selection and Training

- Used TimeGPT (Nixtla)
- Forecast horizon: 639 hours
- Confidence intervals: 80%, 90%
- Baseline model without engineered features



Feature Engineering

- Added features: Hour, Lag temp, Year, Month, Day
- Captured seasonality & trends
- Improved model performance

```
Adding custom features...
Dropped 1 rows with NaNs after creating lag features.
DataFrame with new features:

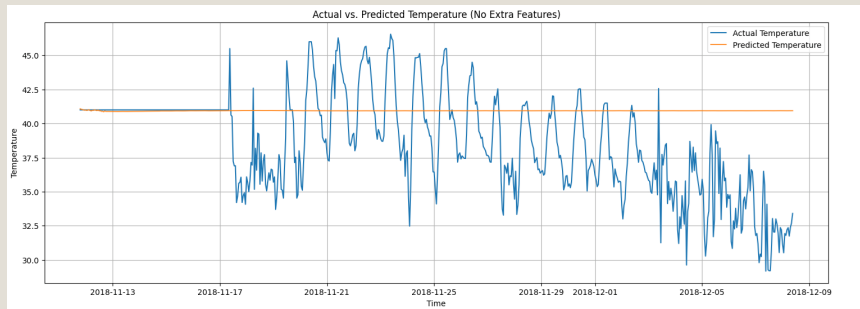
Converted 'noted_date' to datetime index and sorted.
Index type: datetime64[ns]
Data ranges from 2018-07-28 08:00:00 to 2018-12-08 09:00:00
DataFrame head with datetime index:
```

	temp	hour	temp_lag_1	year	month	day
noted_date						
2018-07-28 08:00:00	31.386364	8	31.386364	2018	7	28
2018-07-28 09:00:00	31.386364	9	31.386364	2018	7	28
2018-07-28 10:00:00	31.386364	10	31.386364	2018	7	28
2018-07-28 11:00:00	31.386364	11	31.386364	2018	7	28
2018-07-28 12:00:00	31.386364	12	31.386364	2018	7	28

Model Evaluation

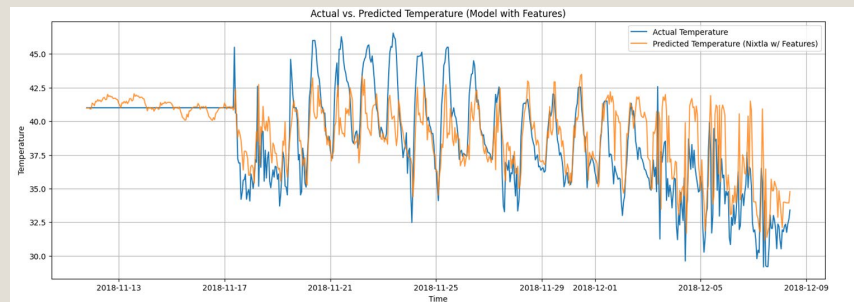
Without Features:

- MAE: 3.2330
- RMSE: 4.2958
- Predictions were not great
- Needed more robust features



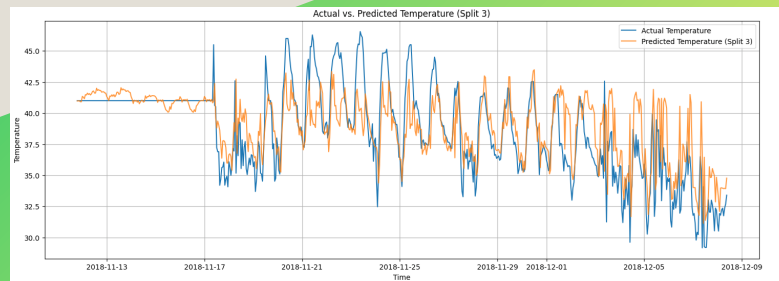
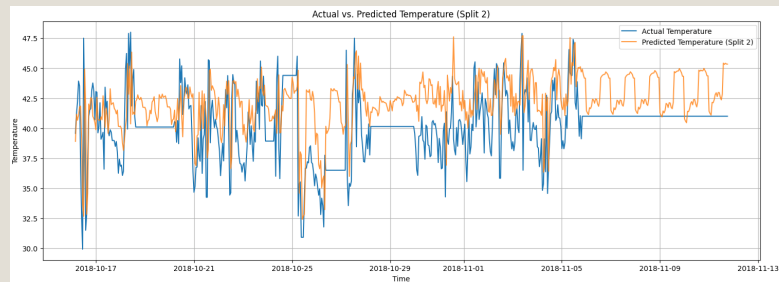
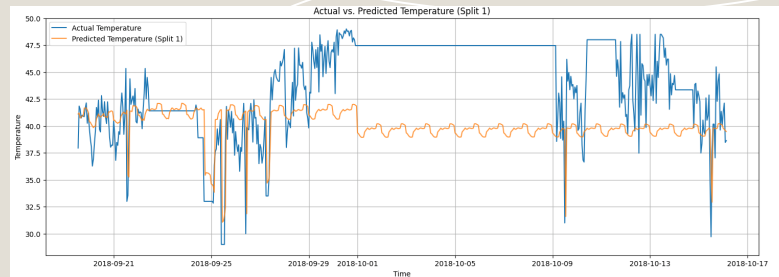
With Features:

- MAE: 1.75
- RMSE: 2.52
- Predictions closely followed actual trends
- Acceptable for IoT use cases



Validation Strategy

- Rolling-Origin Cross Validation
- 3 splits
- Performance improved with more data
- Simulates real-world deployment



Generative Modeling

--- Encoder Summary ---
Model: "encoder"

Layer (type)	Output Shape	Param #	Connected to
input_layer (InputLayer)	(None, 24, 1)	0	-
lstm (LSTM)	(None, 24, 64)	16,896	input_layer[0][0]
lstm_1 (LSTM)	(None, 32)	12,416	lstm[0][0]
dense (Dense)	(None, 16)	528	lstm_1[0][0]
z_mean (Dense)	(None, 8)	136	dense[0][0]
z_log_var (Dense)	(None, 8)	136	dense[0][0]
z (Lambda)	(None, 8)	0	z_mean[0][0], z_log_var[0][0]

Total params: 30,112 (117.62 KB)
Trainable params: 30,112 (117.62 KB)
Non-trainable params: 0 (0.00 B)

--- Decoder Summary ---
Model: "decoder"

Layer (type)	Output Shape	Param #
input_layer_1 (InputLayer)	(None, 8)	0
dense_1 (Dense)	(None, 16)	144
repeat_vector (RepeatVector)	(None, 24, 16)	0
lstm_2 (LSTM)	(None, 24, 32)	6,272
lstm_3 (LSTM)	(None, 24, 64)	24,832
time_distributed (TimeDistributed)	(None, 24, 1)	65

Total params: 31,313 (122.32 KB)
Trainable params: 31,313 (122.32 KB)
Non-trainable params: 0 (0.00 B)

--- VAE Summary (Combined) ---
Model: "vae"

Layer (type)	Output Shape	Param #
encoder (Functional)	((None, 8), (None, 8), (None, 8))	30,112
decoder (Functional)	(None, 24, 1)	31,313

Total params: 61,425 (239.94 KB)
Trainable params: 61,425 (239.94 KB)
Non-trainable params: 0 (0.00 B)

- Used VAE to generate synthetic data
- LSTM-based encoder/decoder
- Generated ~60k data points
- Goal: augment training data

Impact of Generative Models

- Augmentation worsened performance
- MAE increased to 6.46
- Likely due to missing engineered features
- Synthetic data quality matters

```
--- Evaluation Metrics (Model Trained on Augmented Data) ---  
MAE (Augmented): 6.4618  
MSE (Augmented): 53.0503  
RMSE (Augmented): 7.2836  
  
--- Metrics Comparison ---  
Model with Features (No Augmentation): MAE=1.7514, RMSE=2.5173  
Model Trained on Augmented Data: MAE=6.4618, RMSE=7.2836  
Change due to augmentation: MAE +4.7104, RMSE +4.7663  
>>> Augmentation worsened or did not improve MAE
```

Metric	Model with Features (No Augmentation)	Model Trained on Augmented Data	Change (Augmented - Features)
MAE	1.7514	6.4618	+4.7104
MSE	6.3368	53.0503	+46.7135
RMSE	2.5173	7.2836	+4.7663

Challenges, Insights, & Takeaways

- Debugging TimeGPT input issues
- Handling frequency & feature mismatches
- Learned VAE for time series
- Future: better features, tuning, GANs





Thank you